

Machine learning Project – Classification Challenge

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Introduction

Land classification is a time-consuming task when done by hand. Analyzing satellite imagery and manually classifying all pieces of the earth's surface will take an incredible amount of time and labor. Hence, (Geo-Information-) scientists have made efforts to automatize this process, using machine learning algorithms. Though much more efficient, the algorithms bring about their own problems. The high dimensionality of remote sensing data makes classification tasks tricky (Zhao et al., 2023). This is especially true in combination with the limited availability of labelled sample data (Zhao et al., 2023). Furthermore, classification accuracy is often hindered by class imbalance (Wang et al., 2025). Thus, a persistent challenge is selecting the most suitable combination of algorithms, hyperparameters, and accuracy measurement metrics. This project seeks to maximize classification accuracy on Wageningen, using Sentinel-2 data combined with spectral embeddings (Google Earth, 2025). The methodology section will describe how two models were tested and optimized using data from Arnhem and Rhenen. The results section will provide an overview of the main findings. The discussion will contextualize the findings and address strengths as well as weaknesses of this project. Finally, a brief reflection on the ethical, legal and social aspects of this project is provided, by making use of the ELSA scan (Hilten, Mireille van et al., 2024).

Methods

This project aimed to answer one main research question: 'which optimized land cover classification algorithm best succeeds in classifying the land cover of Wageningen, using macro Jaccard as the evaluation metric?' Before fitting the algorithms, an exploratory analysis was conducted using the Red and NIR bands of Sentinel-2, selected based on their discriminative performance observed in Week 1. Red and NIR bands generally work well for distinguishing main land cover classes. A subsample of 10,000 observations (random seed = 42) was used to visualize decision boundaries across multiple classifiers. Linear methods (Logistic Regression, LDA) produced rigid, linear partitions with limited capacity to capture spectral complexity, whereas KNN generated irregular, locally adaptive boundaries. SVM yielded smooth decision regions via its default kernel, while tree-based ensemble methods (Random Forest, Bagging) produced characteristic orthogonal partitions of the spectral space. The plotted decision boundaries as part of the exploratory analysis are shown in figure 1.

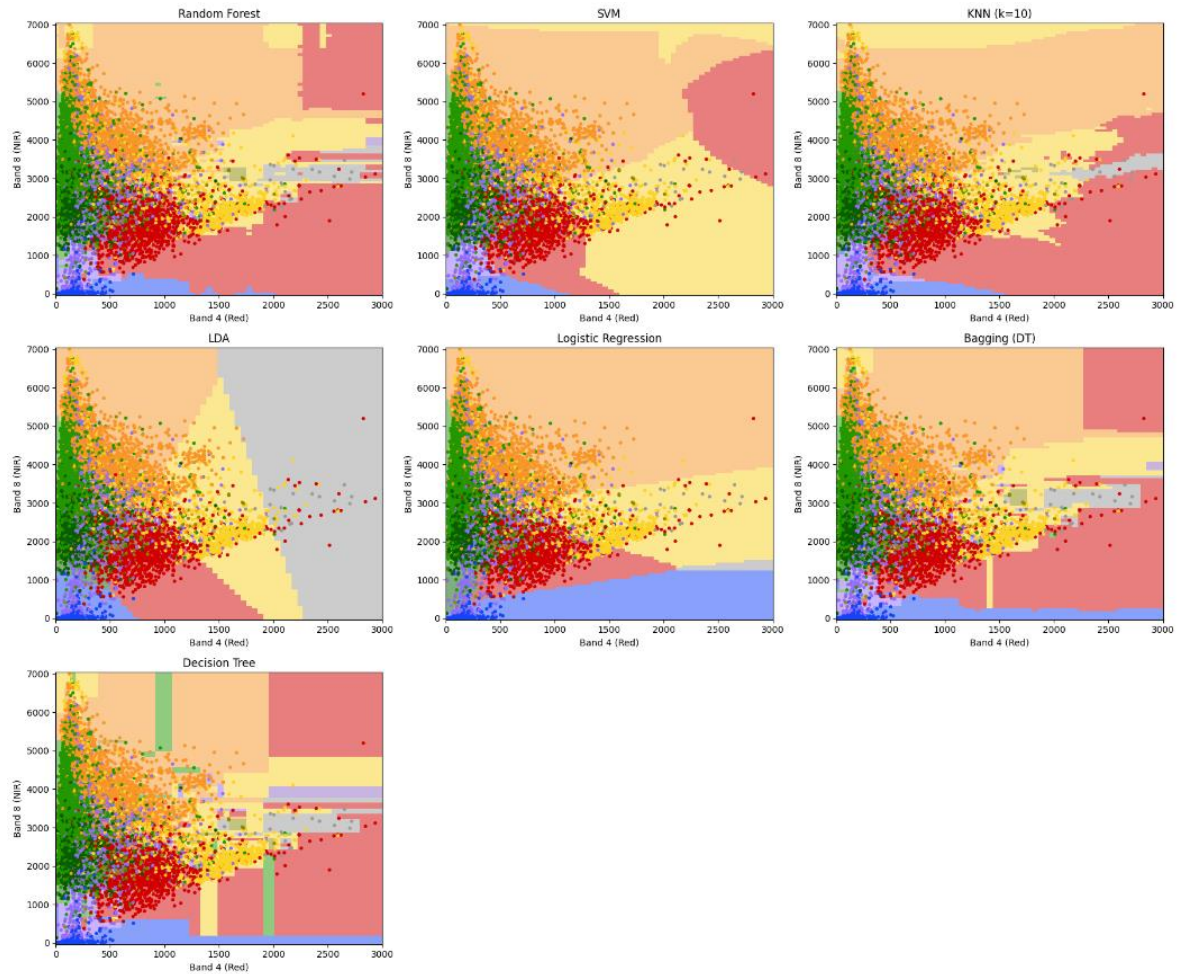


Figure 1: decision boundaries plotted by various algorithms using default settings.

Feature engineering was performed by deriving spectral indices (NDVI, EVI, SAVI, among others) and incorporating Satellite Embeddings V1 (64-dimensional spatially derived latent features), yielding 84 predictors after integration with the original Sentinel-2 reflectance data. The Arnhem and Rhenen datasets were merged into a unified training set to increase sample size, as ensemble methods such as Random Forest and XGBoost exhibit improved generalization capacity with larger training volumes. Although this consolidation eliminates the availability of a dedicated validation subset, maximizing classification accuracy was the primary objective of this exercise. StandardScaler normalization was applied independently to the training and test partitions to mitigate scale-dependent bias across the predictor space.

Random Forest

Due to the capacity for random sample selection (Bootstrapping) and the capacity for random feature selection, which reduces collinearity compared with bagging methods, Random Forest was deemed one of the best models to conduct a classification for non-linear data. Furthermore, because it performs out-of-bag (OOB) estimation, there is no strict necessity of an independent validation set. Additionally, the model used for Landcover maps from Arnhem and Rhenen was created using the Random Forest

algorithm (Malinowski, R., et al, 2020) developing a fully automated classification workflow based on multi-temporal Sentinel-2 imagery, employing a classifier with 50 decision trees trained on samples extracted from existing land cover databases (CORINE Land Cover and High-Resolution Layers), which is a similar workflow for this challenge. In line with the original classification technique used, utilizing the Random Forest model seemed promising. We performed 2 iterations using different hyperparameters, number of features, and training sets to compare the change in accuracy.

Steps:

- 1) Initial approach: training and validation sets with 12 features. Use the default parameters from RF classifier (100 trees). Evaluate the training and validation of accuracy. Create the prediction and upload the csv.
- 2) Second approach: RandomGridsearch using 10.000 random samples (time effort) to find the best hyperparameters: number_trees, max_features and max depth. Select the best parameters and create the final prediction for the image of Wageningen. Compare how the training accuracy and OOB accuracy changes over different iterations of the selected hyperparameters: ntress, max_depth and max_features.

XGBoost

The XGBoost algorithm was originally implemented in an experimental capacity. Both project contributors had no previous knowledge of its operationality. However, literature suggests that XGBoost (or extreme gradient boosting) can outperform other classifiers in multiple scenarios (Dũng, Lương et al., 2025) (Abdi, 2019). This algorithm was selected due to its structural similarities to the Random Forest, as both algorithms are ensemble-based approaches. For this project, XGBoost was implemented as follows:

- 1) The prediction labels (y_train), had to be encoded. The original land cover codes were converted from 58, 68, 76, 77, 95, 96, 98, 99, 113 & 151 to 0-9.
- 2) The HalvingRandomSearchCV algorithm was used to find optimal hyperparameters. This cross-validation method was chosen due to its efficiency. Instead of an exhaustive search, it starts evaluating random hyperparameter combinations and eliminates one third of the candidates each iteration. Considering the extensive datasets, the less computationally heavy cross-validation technique was preferred. Furthermore, for this project, the evaluated hyperparameters were not set too computationally expensive, as the model itself already took some time running.
- 3) The optimal hyperparameters were implemented to compute the final fit. Again, random state of 42 was used. The final fit was then used to predict the land cover labels for Wageningen. The encoded test prediction was transformed back into the original S2GLC labels. The XGBoost algorithm does not perform

bootstrapping. Thus, there was no out-of-bag data, meaning that validation accuracy was also not computed.

Results

Random Forest classification

First RF iteration:

The first model had only 12 features (sentinel spectral bands) and considered only the Arnhem data as training. There was no hyperparameter selection, and the Random forest model considered only default parameters. Due to the minor amount of training samples and the reduced number of features, this model was the fastest of all, with the tradeoff being the least accurate. This iteration achieved an accuracy of 0.43456. One of the mayor findings comparing the macro average accuracy per classes was the clear imbalance between classes were class 5,7 and 8 presented a few thousand of samples compared with the tens of thousands of samples from the other classes. Within this context, it is very likely to obtain a poor performance on the general accuracy due to the class imbalance problem.

Second RF iteration:

The second RF iteration considered 3 mayor changes, the incorporation of more features (84 predictors), the merge of validation and training data sets, and the hyperparameter tuning. The optimal hyperparameters for the model consider the number of trees, the max number of features per tree, and the maximum depth of each tree. The optimally hyperparameters are: 300 trees, 40 max features, and a depth of 10. Obtaining a final accuracy of 0.49, indicating that our model has around a 50% chance to predict the landcover class accurately. This is not particularly a good score for a landcover classification.

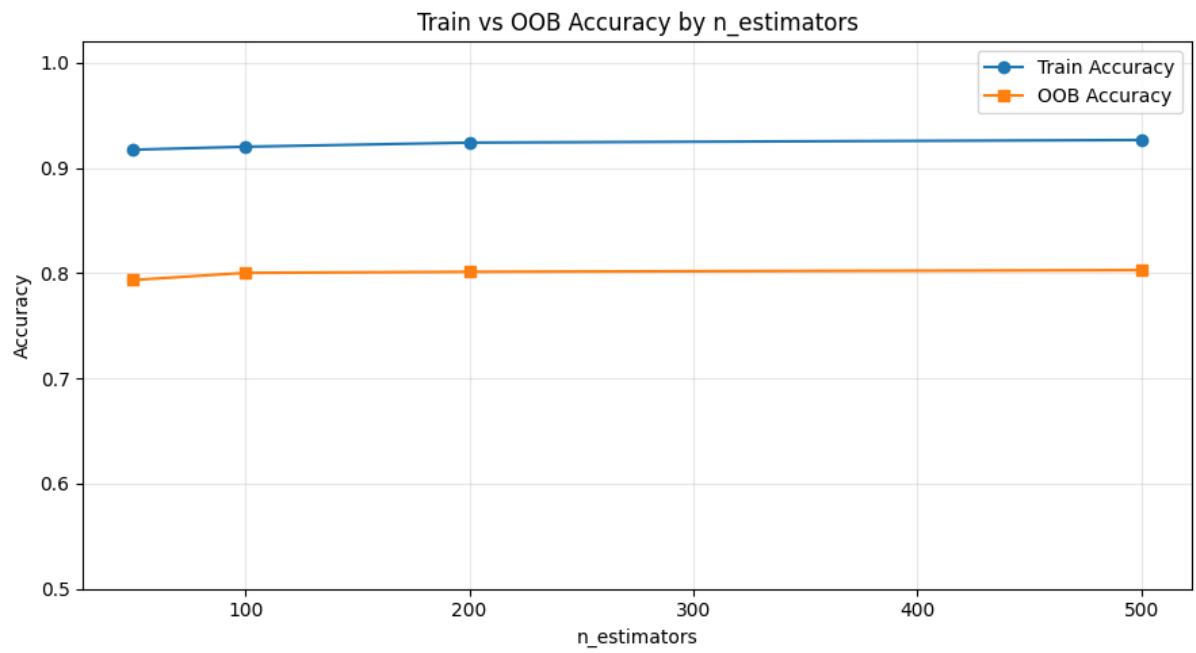


Figure 2: Train and OOB accuracy using different $n_{\text{estimators}}$ for RF

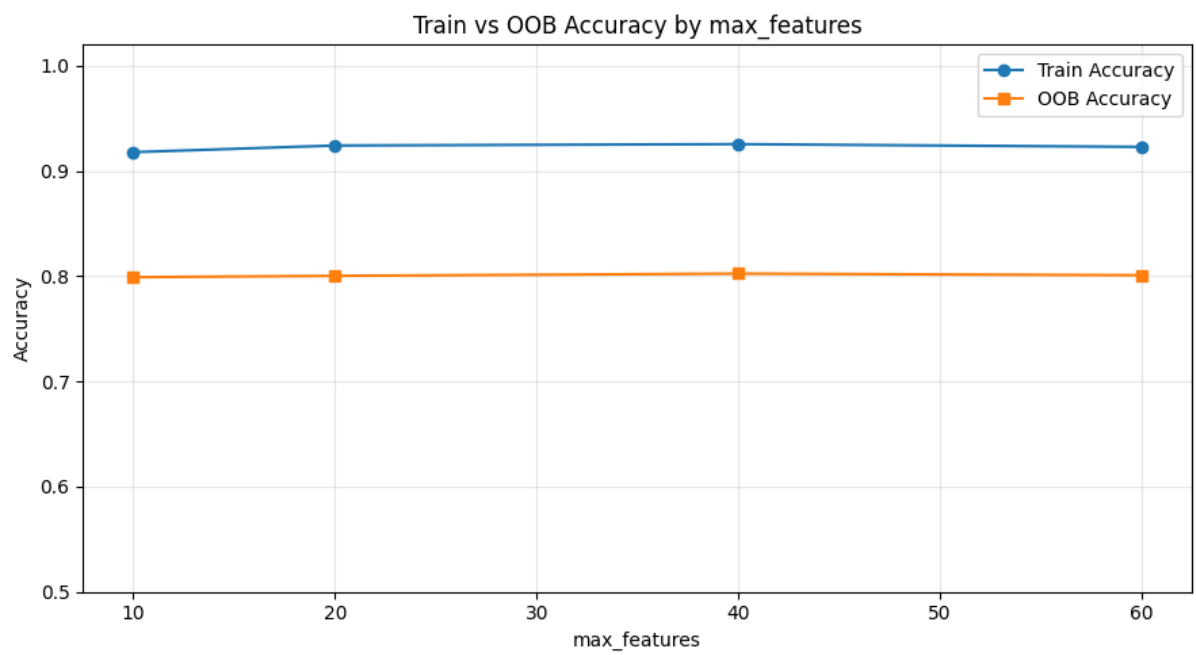


Figure 3: Train and OOB accuracy using different max_features for RF

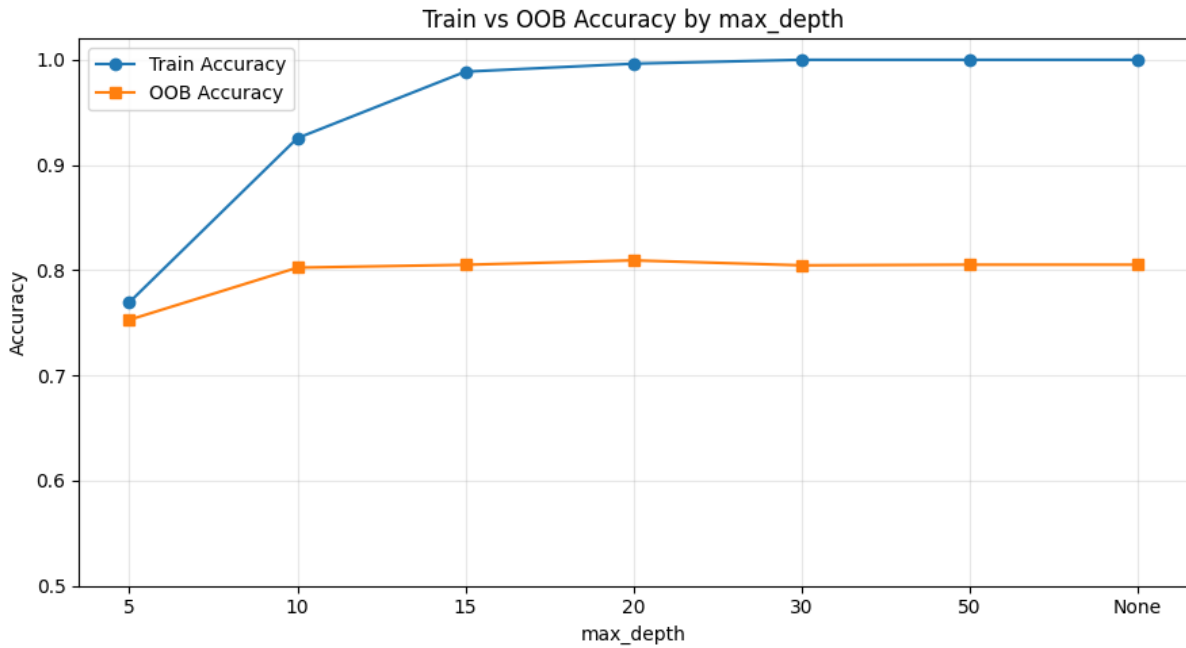


Figure 4: Train and OOB accuracy using different max_depth for RF

Figures 2, 3 and 4 illustrate the divergence between training and OOB accuracy across varying hyperparameter configurations of Random Forest. One of the inherent advantages of RF is its bootstrap aggregation mechanism, which is used here to create an OOB validation set. The scores of this set can be compared against the training set scores. For the number of estimators, the optimal value was found at $n=300$, beyond which accuracy gains are negligible, similar case for the max features selection. For the max depth of each tree, bigger depth tends to overfit the model indicating that grid search did find an optimal depth to reduce this overfitting problem. The gap between training and OOB accuracy reflects the degree of overfitting of the model, indicating the low generalization capacity. Based on this, we could expect poor testing accuracy over the city of Wageningen. Surprisingly small differences were observed between different values for the $n_estimators$ and $max_features$.

Hyperparameter	Value
N_estimators	300
Max_depth	10
Max_features	40

Table 1: chosen hyperparameters for the final Random Forest fit

XGBoost Classification

This model was implemented with the best hyperparameters found by the HalvingRandomSearchCV algorithm. Table 1 contains an overview of the implemented hyperparameters in the final fit.

Hyperparameter	Value
N_estimators	161

Max_depth	3
Learning_rate	0.145
Subsample	0.730
Colsample_bytree	0.940
Min_child_weight	7

Table 2: chosen hyperparameters for the final XGBoost fit

Utilizing these hyperparameters, a test accuracy of 0.50719 was achieved. This is slightly higher than the achieved macro Jaccard score using the Random Forest model with tuned hyperparameters. However, it must be noted that an accuracy of barely 50% is not very high. Possibly, smaller, underperforming classes hurt the accuracy score of the model disproportionately due to the way the Jaccard score is constructed.

Model	Processing time	Accuracy
Random Forest	55 min	0.49
XGBoost	62 min	0.50

Table 3: comparison of the Random Forest and XGBoost models

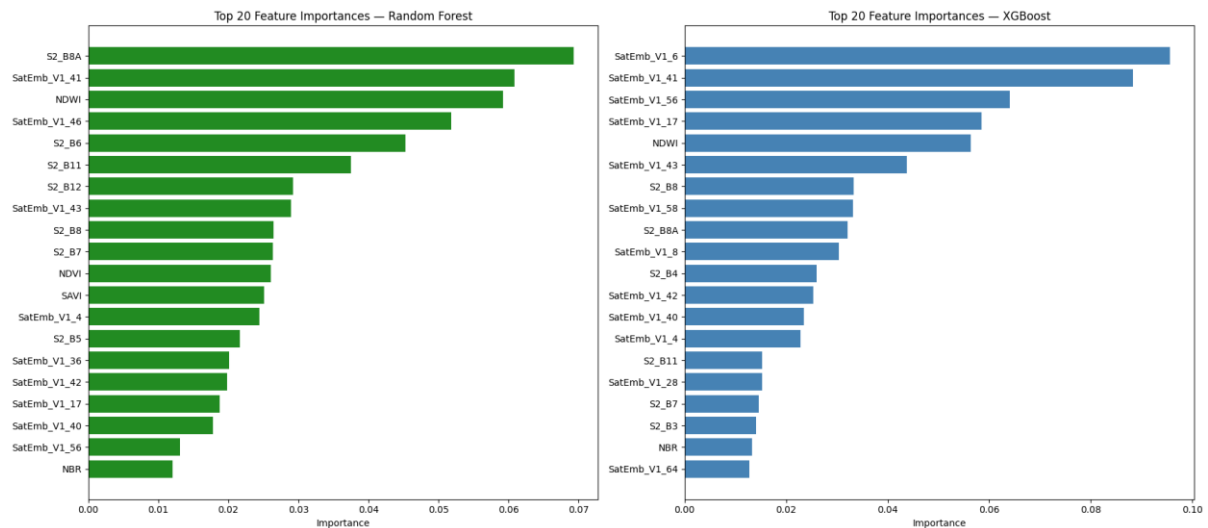


Figure 5: the contribution of the top 20 features to the models' predictions. In green, we see the top 20 features for Random Forest and in blue we see them for XGBoost.

In figure 5, we see the contribution of the top 20 features to the models' predictions. A discrepancy between the features used by both models can be observed. Random Forest relied more on the created indices and spectral bands while the XGBoost model made stronger use of spatial embeddings. The heavy contribution of spatial embeddings arguably makes the models, especially the XGBoost model, hard to interpret.

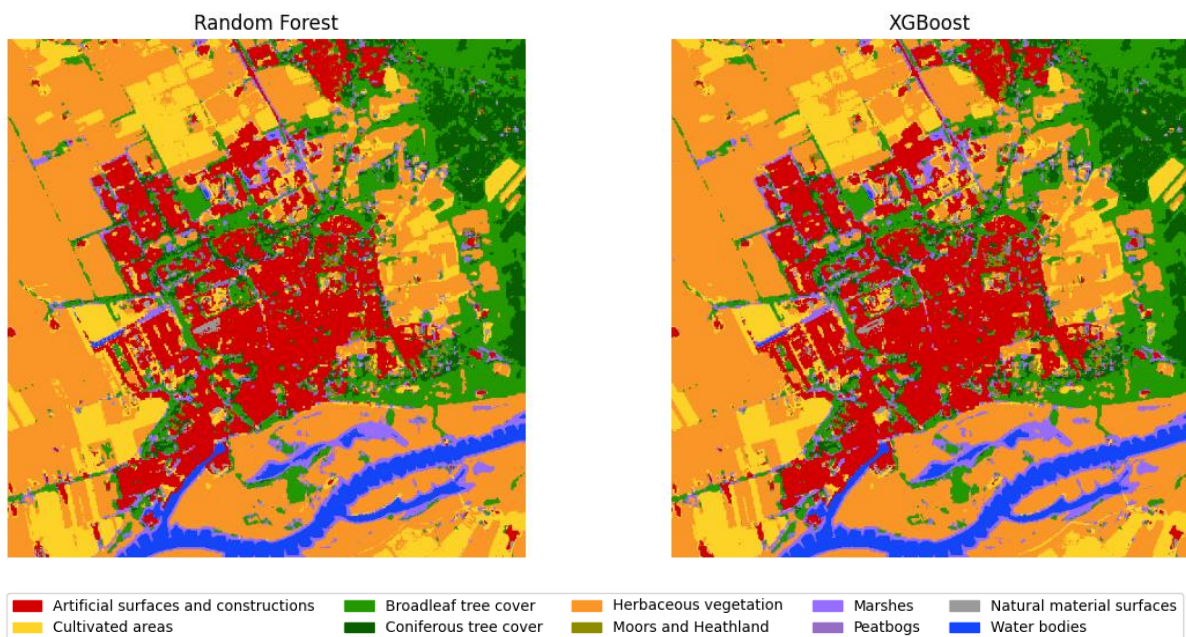


Figure 6: predicted classification maps for Wageningen, performed by the Random Forest model on the left and the XGBoost model on the right.

In figure 6, we see the predicted classification maps for Wageningen, performed by the Random Forest model and the XGboost model. It is hard to distinguish between the two maps. This is not surprising, as the accuracy score for both models was very similar. No major classification difference can be observed, which could point to the fact that both models have similar patterns in misclassification.

Discussion

During this challenge we were encouraged to obtain the best test accuracy possible for a landcover classification problem, where some assumptions of the validation and training data sets were violated to increase accuracy. Because Random Forest can perform an "internal cross-validation", the merge of data sets is not much of a problem but an advantage of this model. Including geographical data from different cities can improve the capacity of the model to generalize with new unseen data like Wageningen. Even though RF didn't show a good performance in terms of general accuracy of 49%. Based on the accuracy of the model with different hyperparameter tuning, we can conclude that the number of trees and max features barely changes the final OOB accuracy, which is a potential estimator of the test accuracy. Nevertheless, the max depth hyperparameter showed important changes in the training accuracy where high parameter values will increase overfitting. Even so, the model shows an overfitting tendency if we compare the overall training and OOB accuracy for all hyperparameters in RF. Answering the research question; the XGBoost model performs best when considering the macro Jaccard accuracy score. However, this model only slightly outperforms the Random Forest model, while being more computationally costly. When strictly looking for optimization, the XGBoost emerges as the best model. When we consider the balance between computation time and achieved accuracy, it could be argued that the Random Forest model is a more suitable choice. Due to the significant class imbalance in the number of samples, the classification results are unlikely to improve substantially beyond this point. Even applying stratified sampling or upsampling techniques would be very unlikely to change the final score, given the inherent underrepresentation of these classes across the city of Wageningen. Finally, the use of post-classification filters to reduce the salt-pepper effect over heterogeneous areas like the city center of Wageningen or roads with different vegetation landcover types, could improve the final accuracy of the model. Class aggragation is another solution to the small classes over the area that could improve the macro Jaccard accuracy.

Strengths

This project's approach has several strong attributes. The most significant strength of our approach was the addition of AlphaEarth's spatial embeddings. This increased our number of features to 84. More importantly, these features include spatial context by encoding information from the surrounding pixel neighborhood (Google Earth, 2025). This has resulted in a significant increase in testing accuracy. This falls in line with existing literature, as the spatial embeddings have been designed to solve classification problems (Google Earth, 2025). Regarding the accuracy metric, Macro Jaccard forces the model to perform well on rare classes as well, instead of just on dominant ones.

As discussed, both the Random Forest model and the XGBoost model have their own strengths. The XGBoost model is robust to correlated features, which makes it very useful

to classify using indices, bands and embeddings. Each tree learns from previously made mistakes, making it more potent than a Random Forest. However, Random Forest performs bootstrapping, creating its own out-of-bag validation set. Another major strength of the Random Forest model is that it computes its trees simultaneously instead of sequentially. This makes the Random Forest model the computationally cheaper option. In this project, the computation time of both models is rather close. However, comparing table 1 and 2, we see that the optimal `n_estimators` and the `max_depth` are set lower for XGBoost than for Random Forest. If this were not the case, it is expected that XGBoost would have a much longer runtime.

Disadvantages/Limitations

Land cover mapping remains a challenging task that involves an iterative process of trial and improvement. Working with multiple land cover classes is particularly difficult due to their spectral similarity — classes that share similar reflectance patterns tend to be confused by the classifier, increasing misclassification rates. This issue is further compounded by class imbalance, where certain land cover types dominate the study area while others are underrepresented, ultimately affecting the overall accuracy of the model. This is further amplified by the macro Jaccard accuracy metric, which treats all classes as equally important. This results in underperforming classes disproportionately hurting the accuracy score. This is a specific restriction over the challenge because we knew that the original LULC maps were made for Europe where these unrepresented classes could be more represented in the whole continent. This is exactly one of the possible limitations of this work. Furthermore, while it increases the training size, the merging of the Rhenen and Arnhem sets means that there's no real validation set. As discussed, this is mainly a limitation for the XGBoost model. Finally, the inclusion of spatial embeddings reduces the interpretability of the project. The construction process of spatial embeddings is a 'black box', meaning that it is unsure what the embeddings mean or represent.

ELSA reflection

From an ethical perspective, our land cover classification model uses publicly available Sentinel-2 satellite imagery, which does not capture individual-level data, reducing privacy concerns. However, misclassification of land cover could have real consequences if used for policy decisions such as urban planning or agricultural subsidies. A model with a 49 – 50% accuracy score could lead to incorrect resource allocation, disproportionately affecting communities in misrepresented areas. Stakeholders must be clearly informed of this major limitation before acting on the produced output. An ethical strength of this project is the use of the macro Jaccard score. This score ensures fairness across all land cover types by equally weighing their importance. This way, rare classes are not ignored in the accuracy assessment. This is essential in, for example, conservation contexts. The feature-importance scores

provided for our models are important regarding transparency. However, many of these features are spatial embeddings, which act as black boxes. This significantly limits the interpretability and transparency of our models to stakeholders. From a legal perspective, all data used is open source. The open-source nature can spark a question of responsibility: which stakeholders are (held) responsible when wrongful classification leads to faulty decision making? Between the developers of the model, the providers of the underlying embeddings and the policy makers, it is unsure who is liable. Socially, algorithms could lower the technical standards for land classification appliances. However, most ground truth labels will still be provided by experts.

Conclusion

After tuning and comparing both selected methods, this project leads to some key findings. The inclusion of more features increments the final accuracy, indicating that the spectral richness of Sentinel-2 and the satellite embeddings are key for distinguishing land cover classes, and that the model benefits more from additional spectral information than from complex hyperparameter tuning. This suggests that future improvements should focus on feature extraction rather than model complexity. However, it is key to keep in mind that model and feature selection come with their own disadvantages. Complications arise due to the differences in algorithm structures and the decrease of interpretability with an increase in features. In our case, we have found that XGBoost slightly outperforms Random Forest. However, it can be argued that Random Forest provides a better balance between accuracy and computation time. From an ethical point of view, algorithms and open-source data usage creates both opportunities as well as complications.

Appendix

Iteration	Model	Parameters	Parater tuning runtime (minutes)	Runtime (minutes)	Accuracy (Macro Jaccard)	Note
1	Random Forest V1	Default (n_estimators=100)	N.A.	1 minutes	0.4345	Default model.
2	Random Forest V2	Optimized (see table 1)	9 minutes	46 minutes	0.49695	Merged validation and training set, included spectral embeddings.
3	XGBoost V1	Default	N.A.	17 minutes	0.47405	Merged validation and training set, included spectral embeddings.
4	XGBoost V2	Optimized (see table 2)	1 hour	2 minutes	0.50719	Merged validation and training set, included spectral embeddings.

Table A1: overview of all iterations in our notebooks and their attributes

Sources

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